

CS:5980

Foundations of Embedded Systems

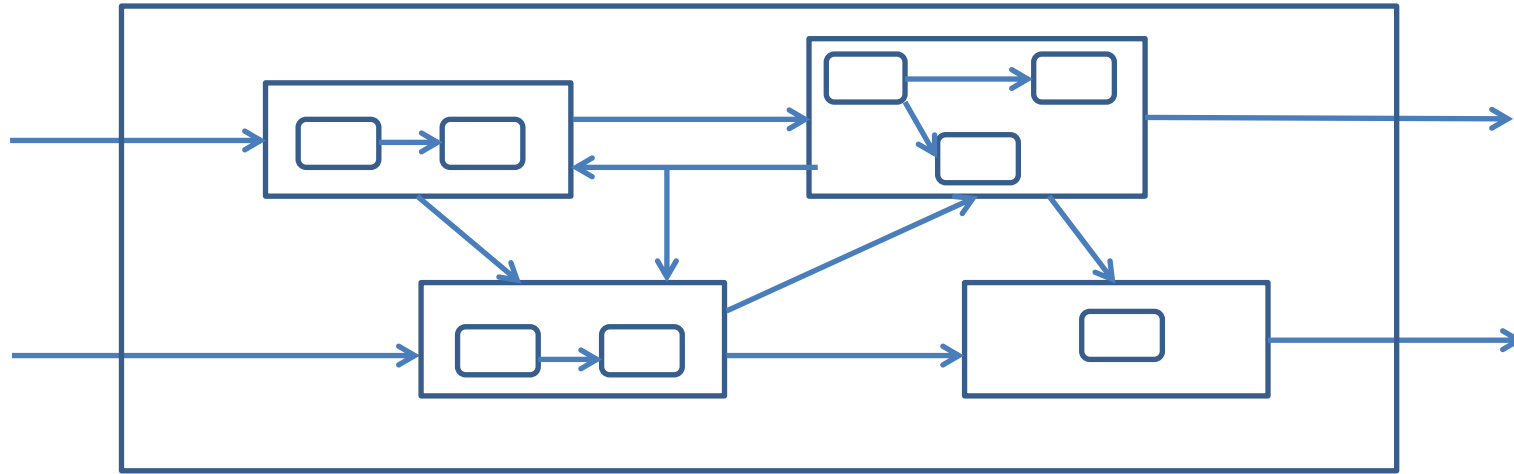
The Synchronous Model

Part III

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Synchronous Design



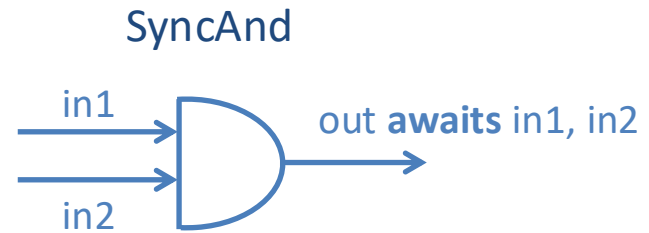
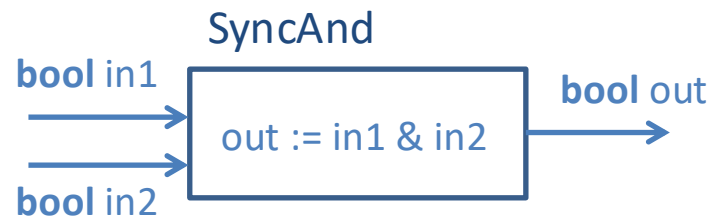
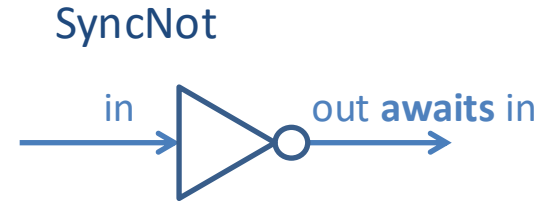
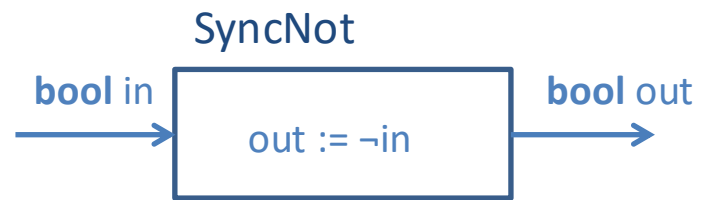
Component (de)composition can be stratified

Bottom-Up Design

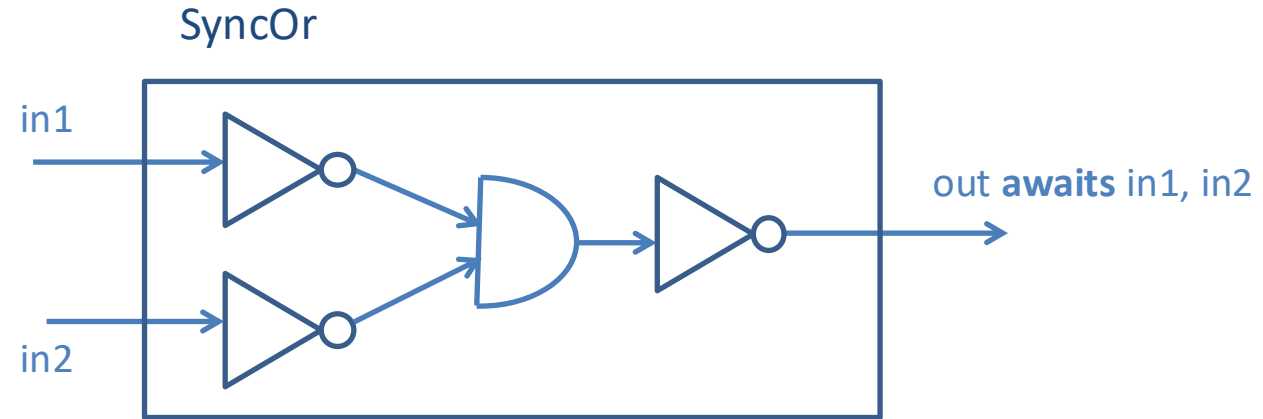
- Design basic components
- Compose existing components in block-diagrams to build new components
- Maintain a library of components, and try to reuse at every step

Canonical example: Synchronous circuits

Combinational Circuits

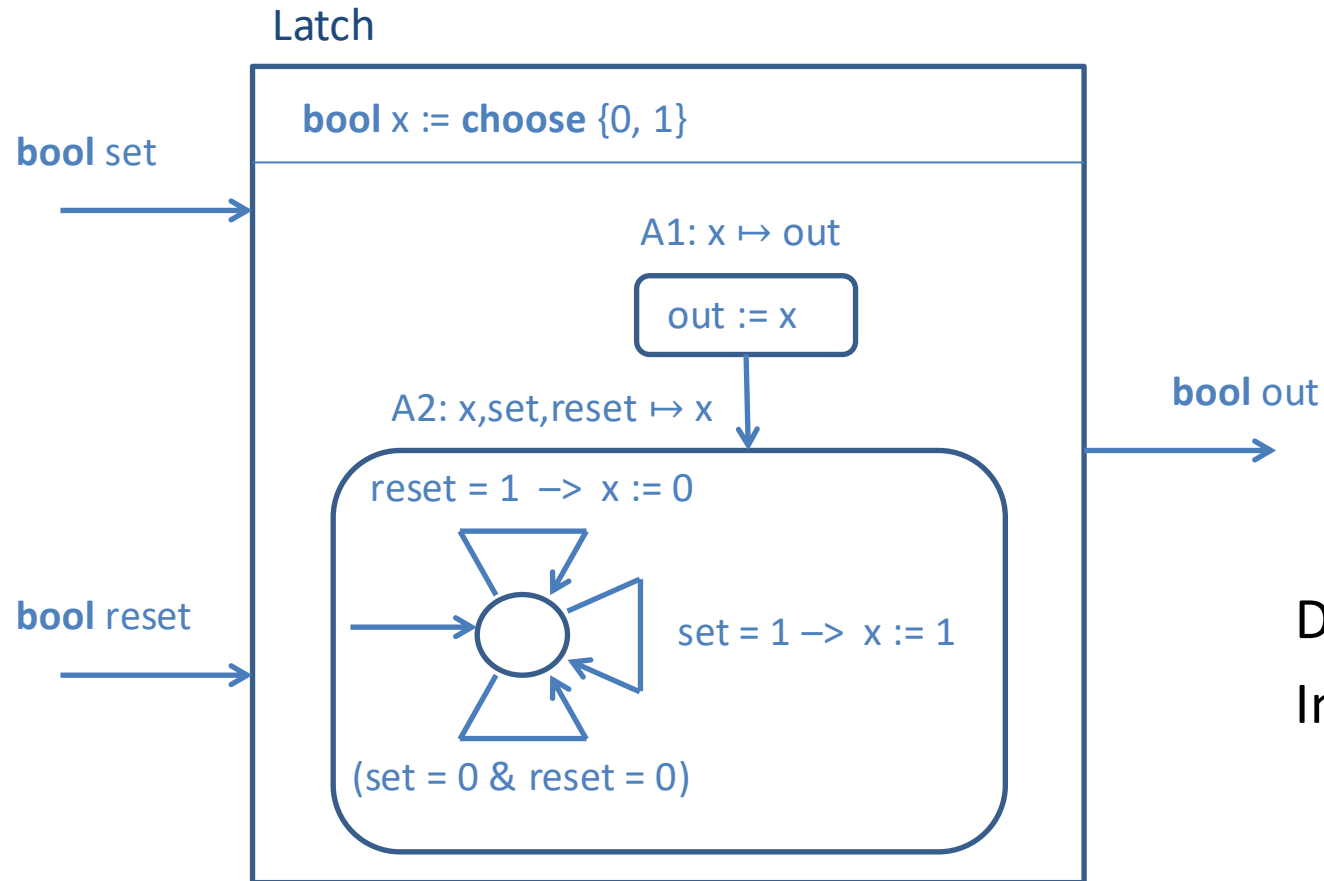


Design OR gate



Recall: $A \vee B \equiv \neg(\neg A \wedge \neg B)$

Synchronous Latch

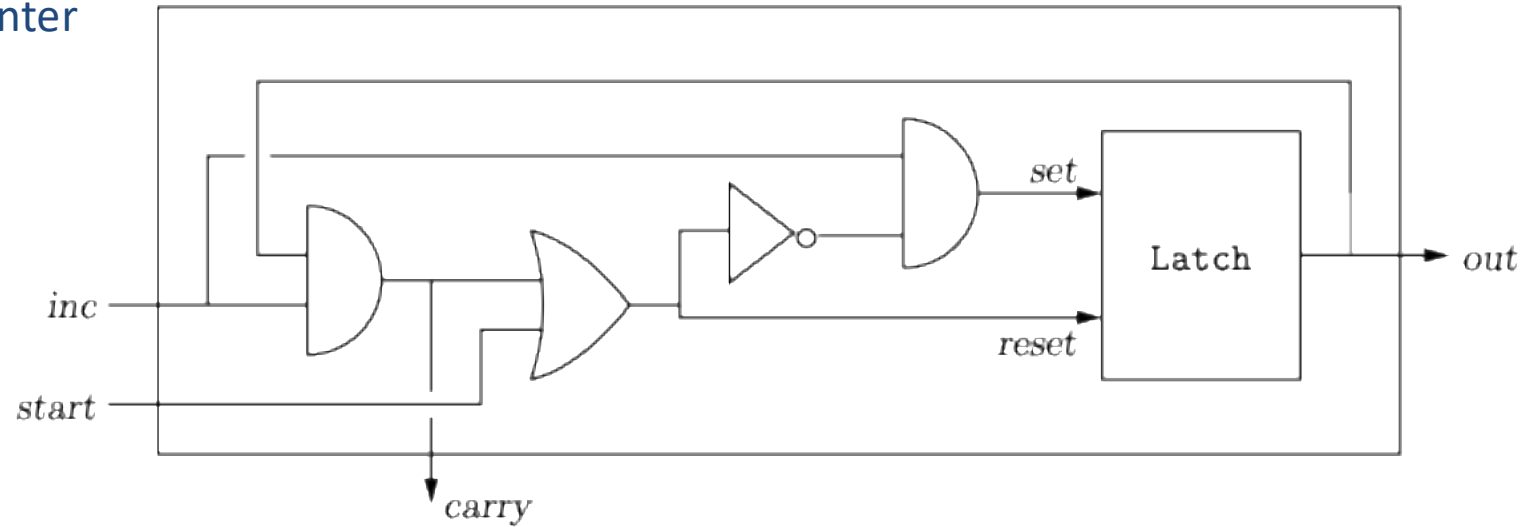


Deterministic?
Input-enabled?

Stores a single bit of information and holds it until it is updated by new input signals

Designing Counter Circuit (1)

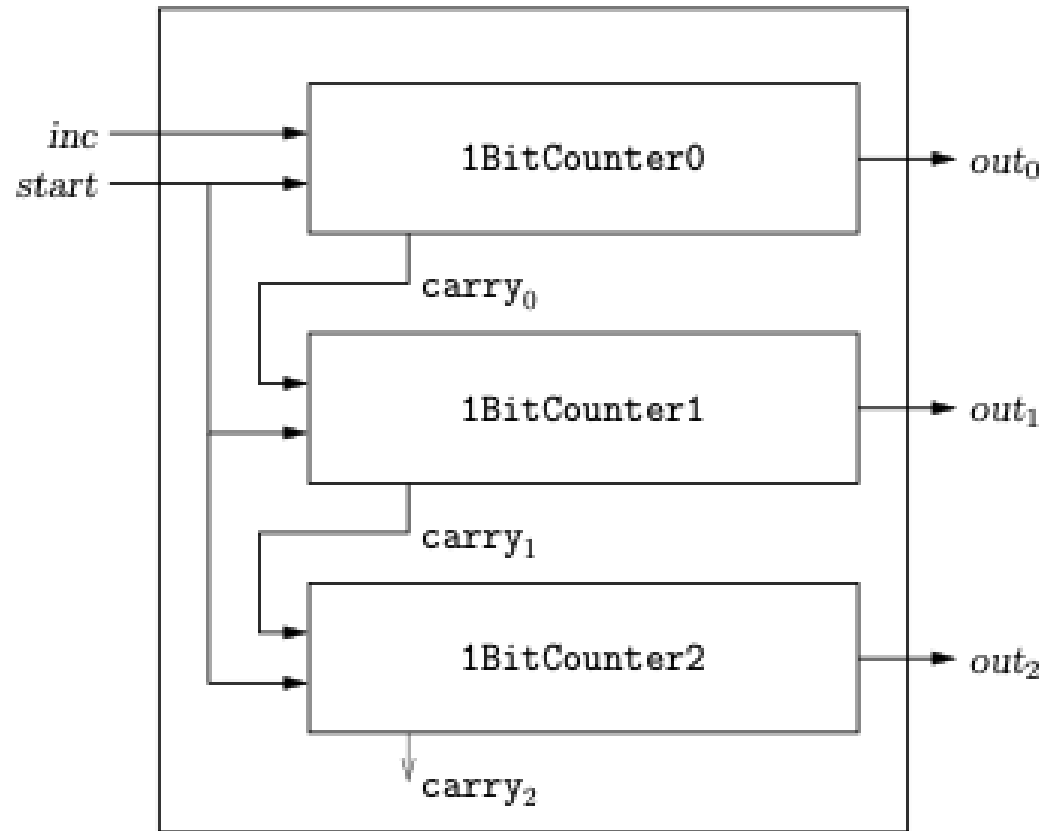
1BitCounter



Are await-dependencies acyclic?

Designing Counter Circuit (2)

3BitCounter

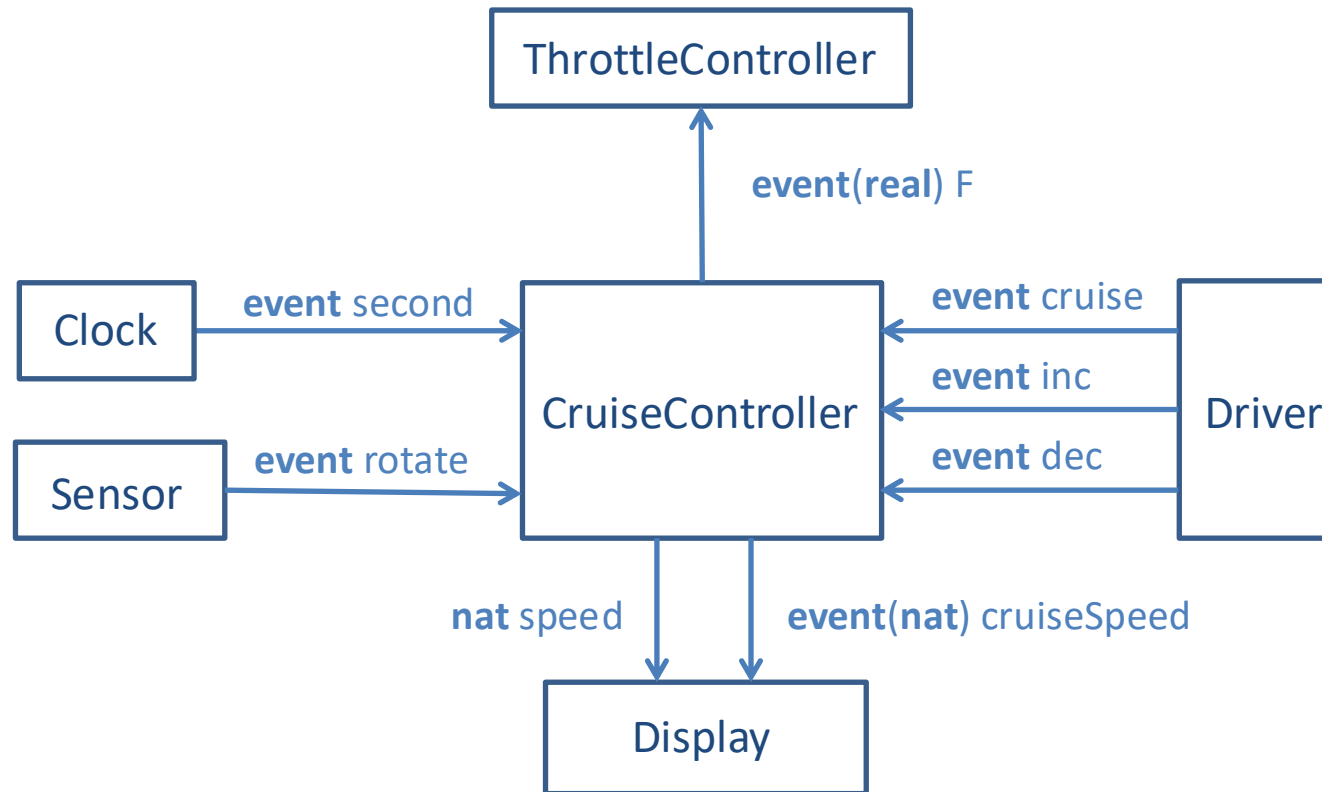


Top-Down Design

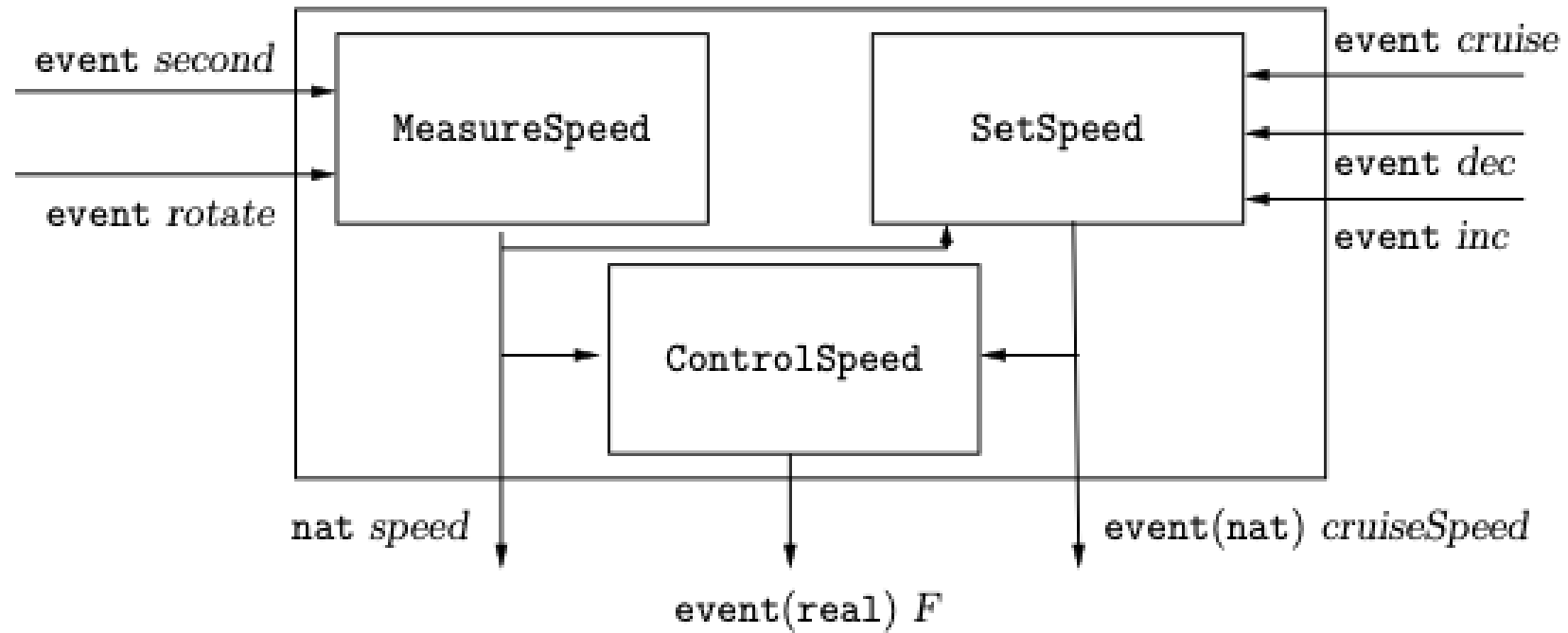
- **Starting point:** Inputs and outputs of desired system C
- Models/assumptions about the environment C operates in
- Informal/formal description of C's desired behavior

Example: Cruise Controller

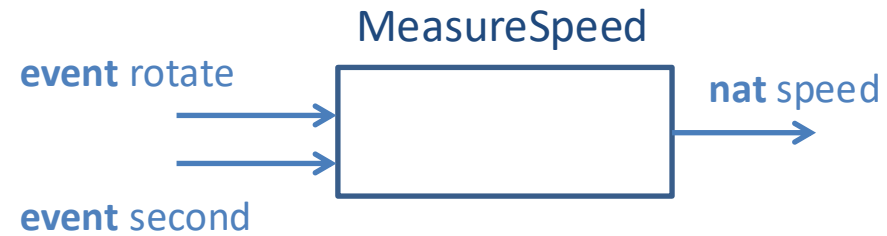
Top-Down Design of a Cruise Controller



Decomposing CruiseController



Tracking Speed

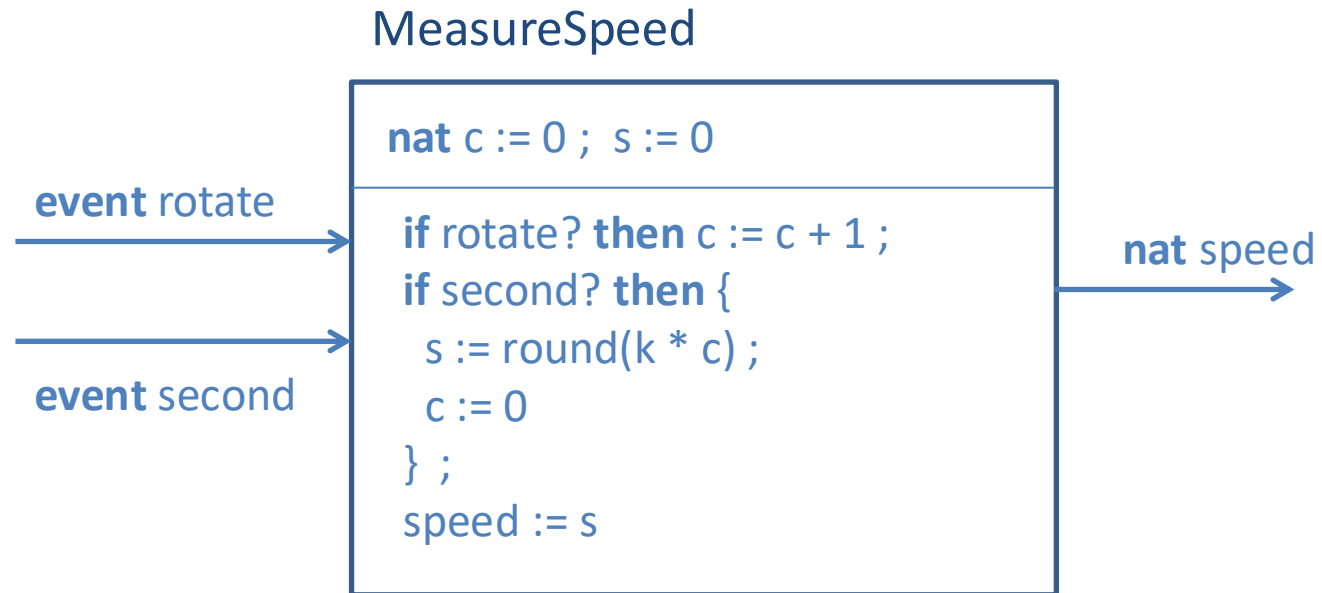


Inputs: Events **rotate** and **second**

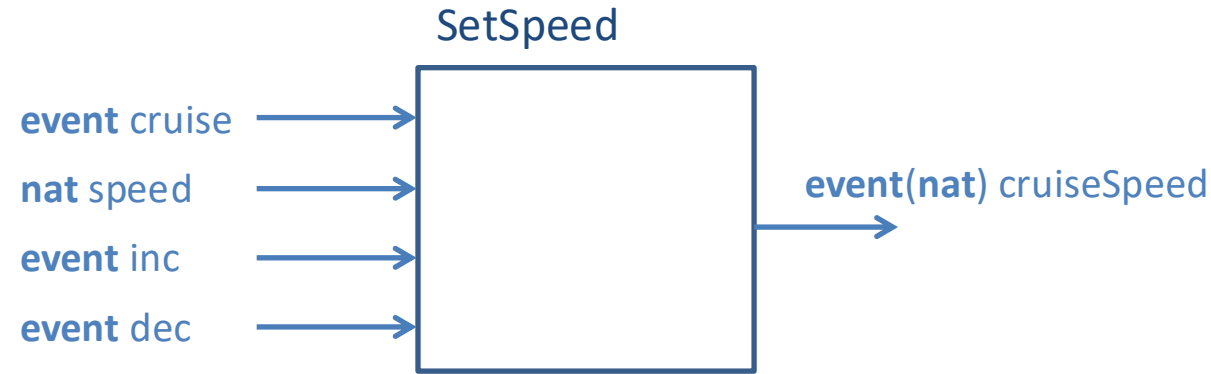
Output: current **speed**

Behavior: Computes the number of rotate events per second

Tracking Speed



Tracking Cruise Settings



Inputs from driver: commands to turn the cruise-control **on/off** and **increment/decrement** desired cruising speed from driver

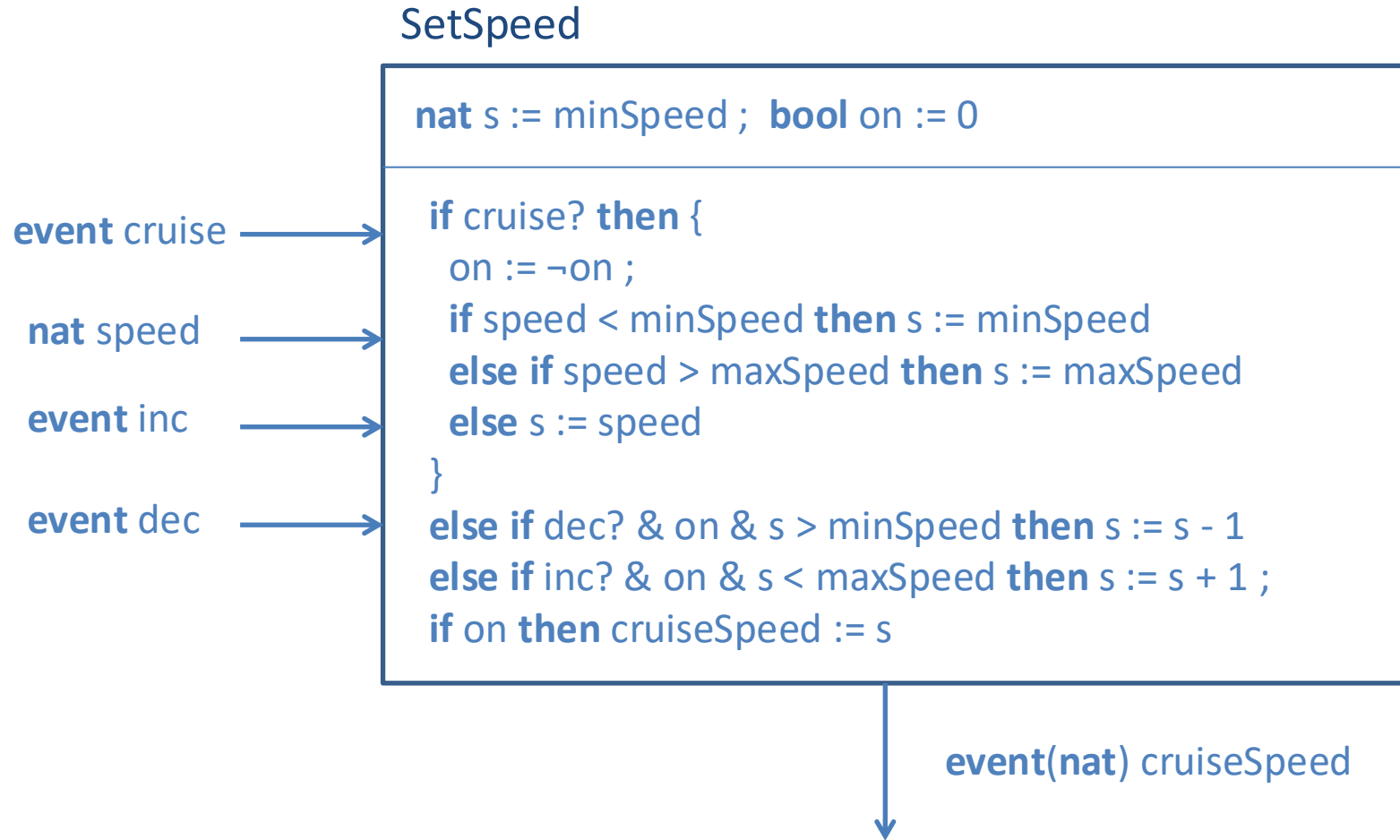
Input from MeasureSpeed: **current** speed

Output: desired **cruising** speed

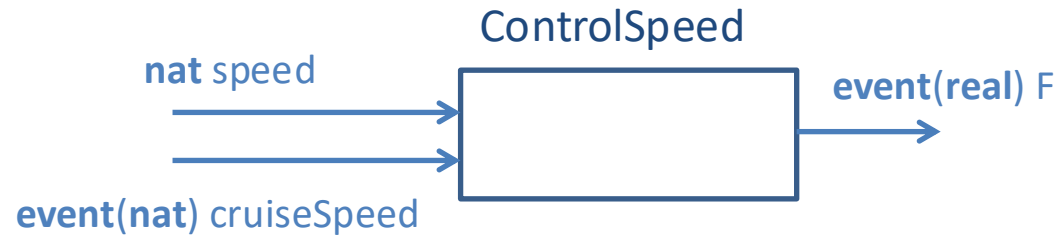
What **assumptions** can we make about simultaneity of events?

Should we include **safety checks** to keep desired speed within bounds?

Tracking Cruise Settings



Controlling Speed



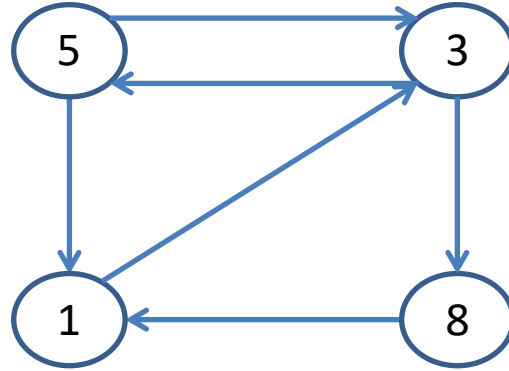
Inputs: Actual speed and desired speed

Output: Pressure on the throttle

Goal: Make actual speed equal to the desired speed
while maintaining key physical properties such as stability

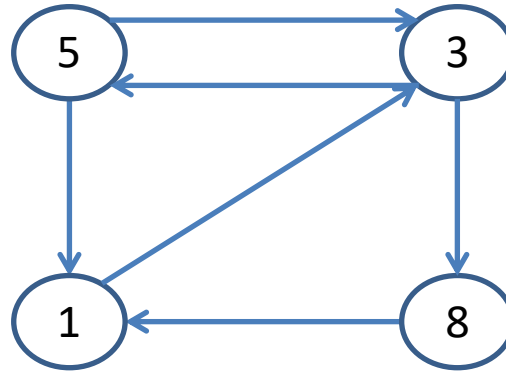
Design relies on theory of dynamical systems (Chapter 6)

Synchronous Networks



- Time divided into slots, with all nodes **synchronized**
- In one round, each node can get a message from each neighbor
- Design abstraction for simplicity
- Some implementation platforms directly support such a **time-triggered** network: WirelessHART (control), CAN (automotive)

Modeling Synchronous Networks



Assumption: Each **link** is **directed** and connects two nodes

Alternative: Broadcast communication (everyone can listen)

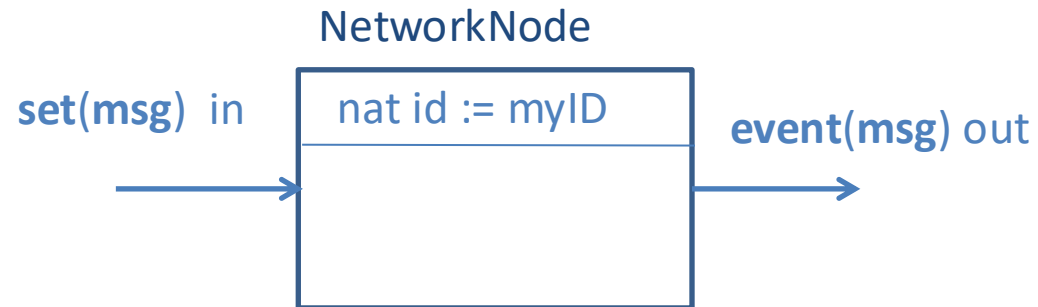
Assumption : Communication is **reliable**

Alternative: Messages may be lost, collisions in broadcast

Network is a **directed graph**

Each link can carry **one message** in **each time** slot

Component for a Network Node



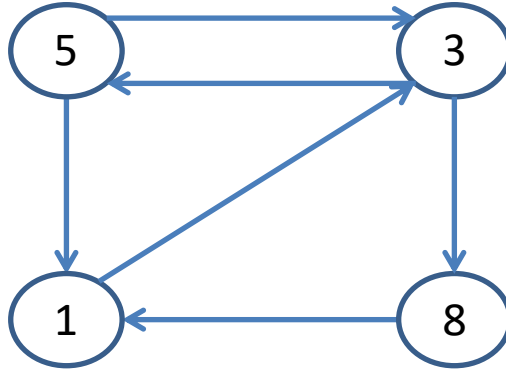
A node does not know network topology

- Each node has **unique identifier**, `myID`
- Does not know which nodes it is connected to
- Useful for **network identification** problems

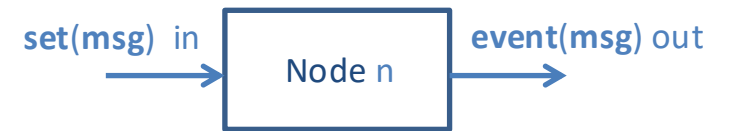
Interface for each node:

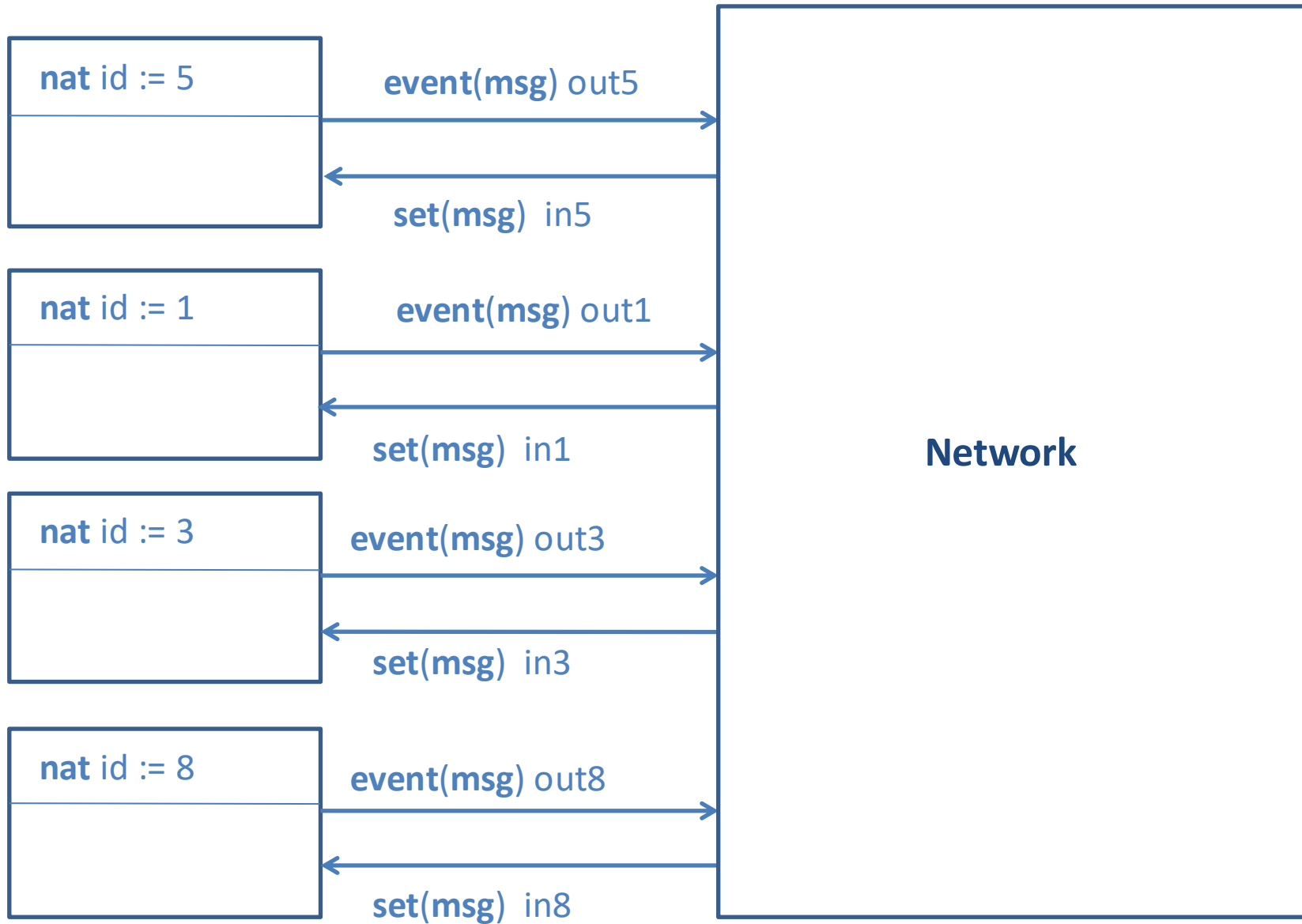
- **Input:** a set of `msg` (delivered by the network)
- **Output:** an event carrying `msg` (may be absent in some rounds)
- Output should not await input

Modeling Synchronous Networks

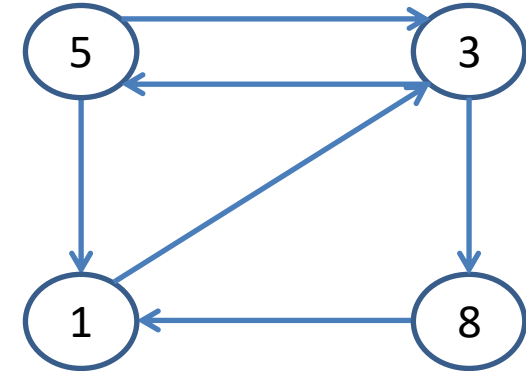
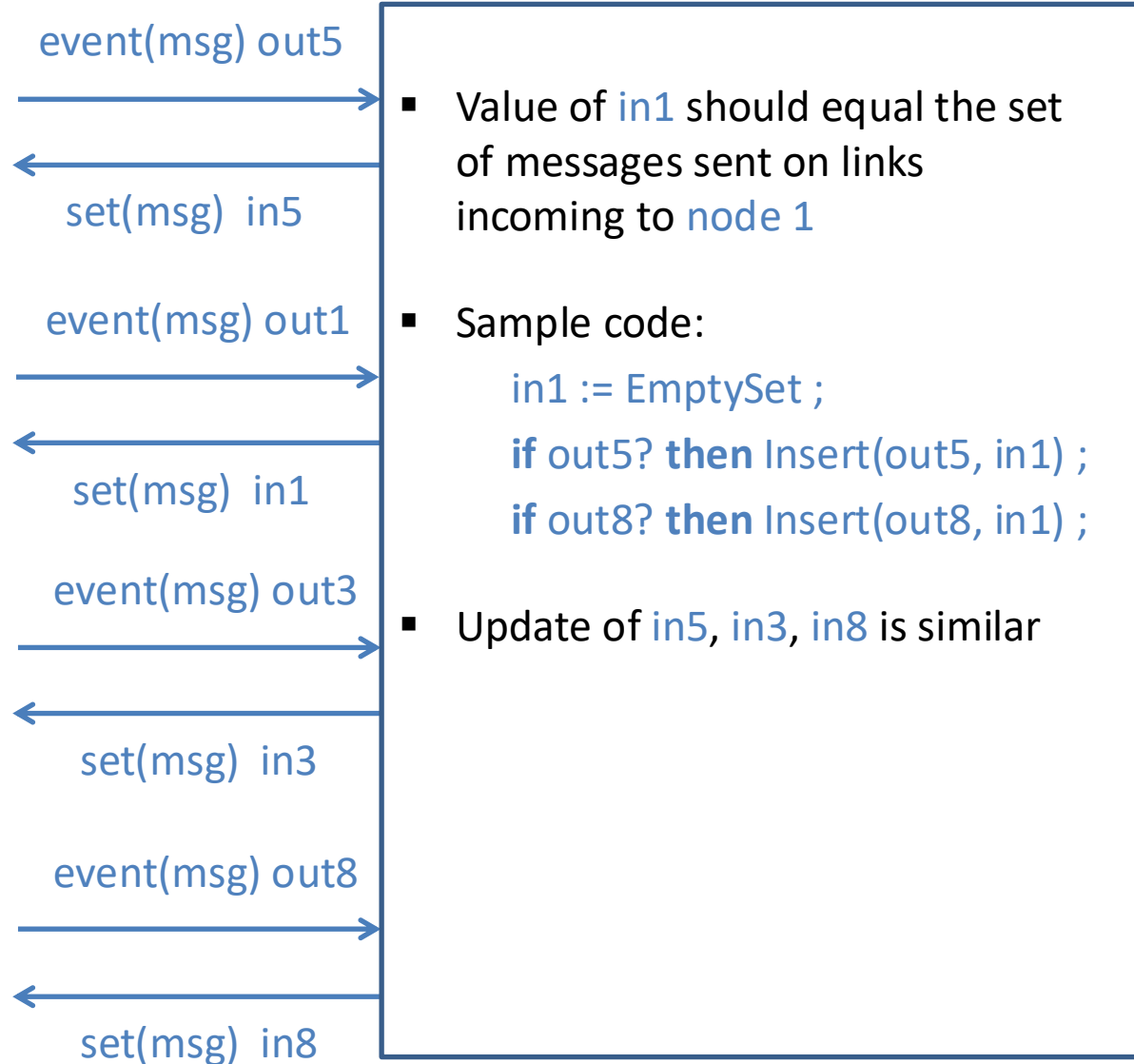


- The description of each **node** is **independent from the network**
- The **network** itself is modeled as a **synchronous component**
- The network description depends on the network graph
 - **Input variables:** for each node n , in_n of type **set(msg)**
 - **Output variables:** for each node n , out_n of type **event(msg)**
- Network is a combinational component
(simply routes messages)

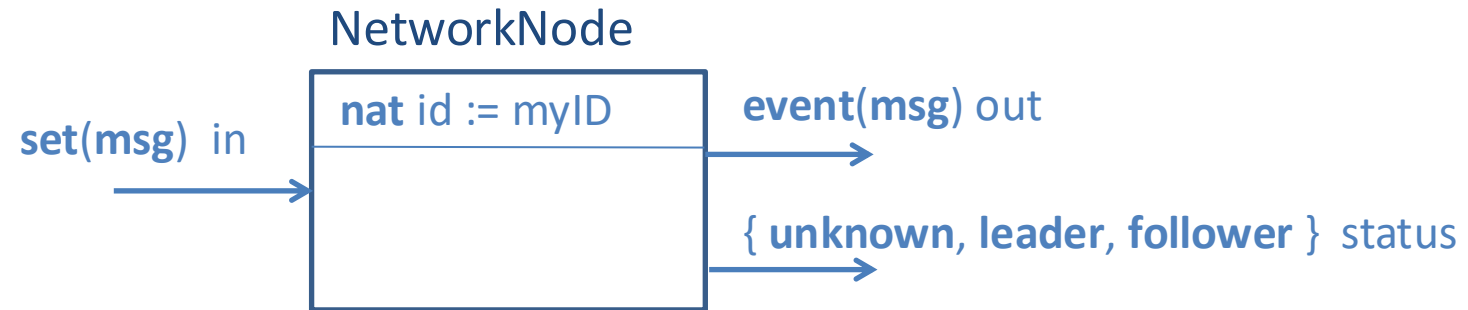




Network



Leader Election Protocol



Classical coordination problem: Elect a **unique** node as a **leader**

- Exchange messages to **find out** which **nodes** are **in network**
- **Output** the **decision** using the variable **status**

Requirements:

1. **Eventually every node** sets status to either **leader** or **follower**
2. **Only one** node sets its status to **leader**

Leader Election: Flooding Algorithm

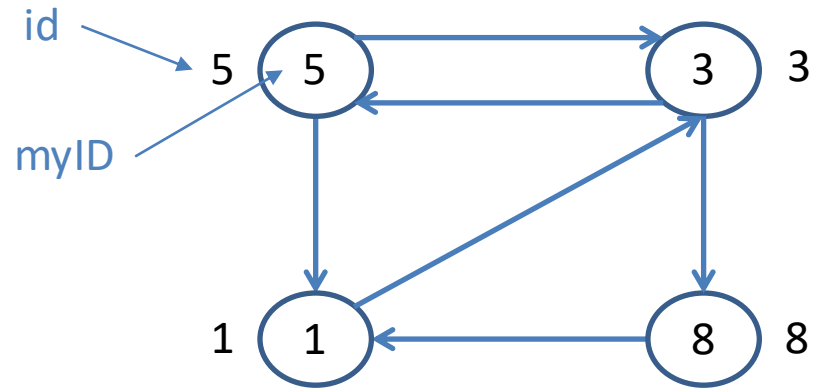
Goal: Elect the node with **highest identifier** as the **leader**

Strategy: Transmit to your neighbors the **highest id** you have **encountered so far**

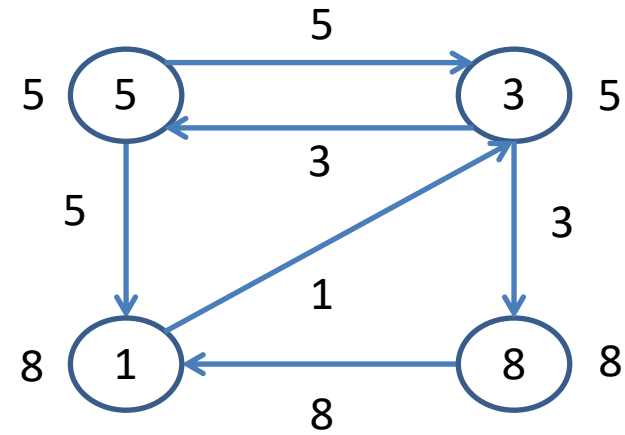
Implementation:

- Maintain a state variable, **id**, initialized to your own identifier
- In each round,
 1. transmit value of **id** on output
 2. receive input values from the network
 3. if a value higher than **id** received, then update **id**

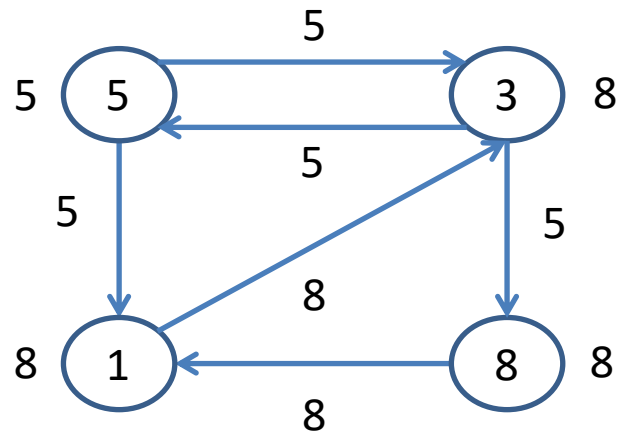
Execution of Leader Election



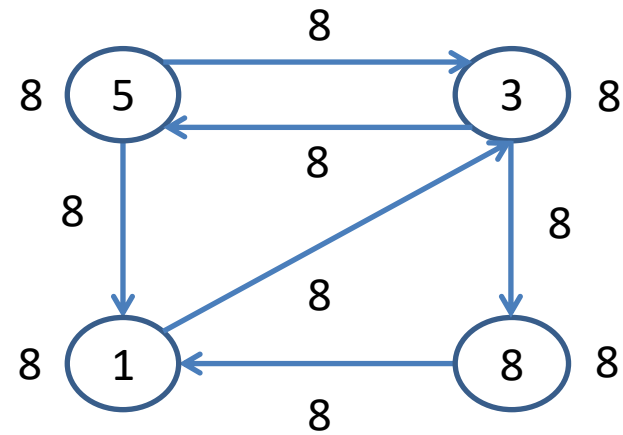
Initially



First round



Second round

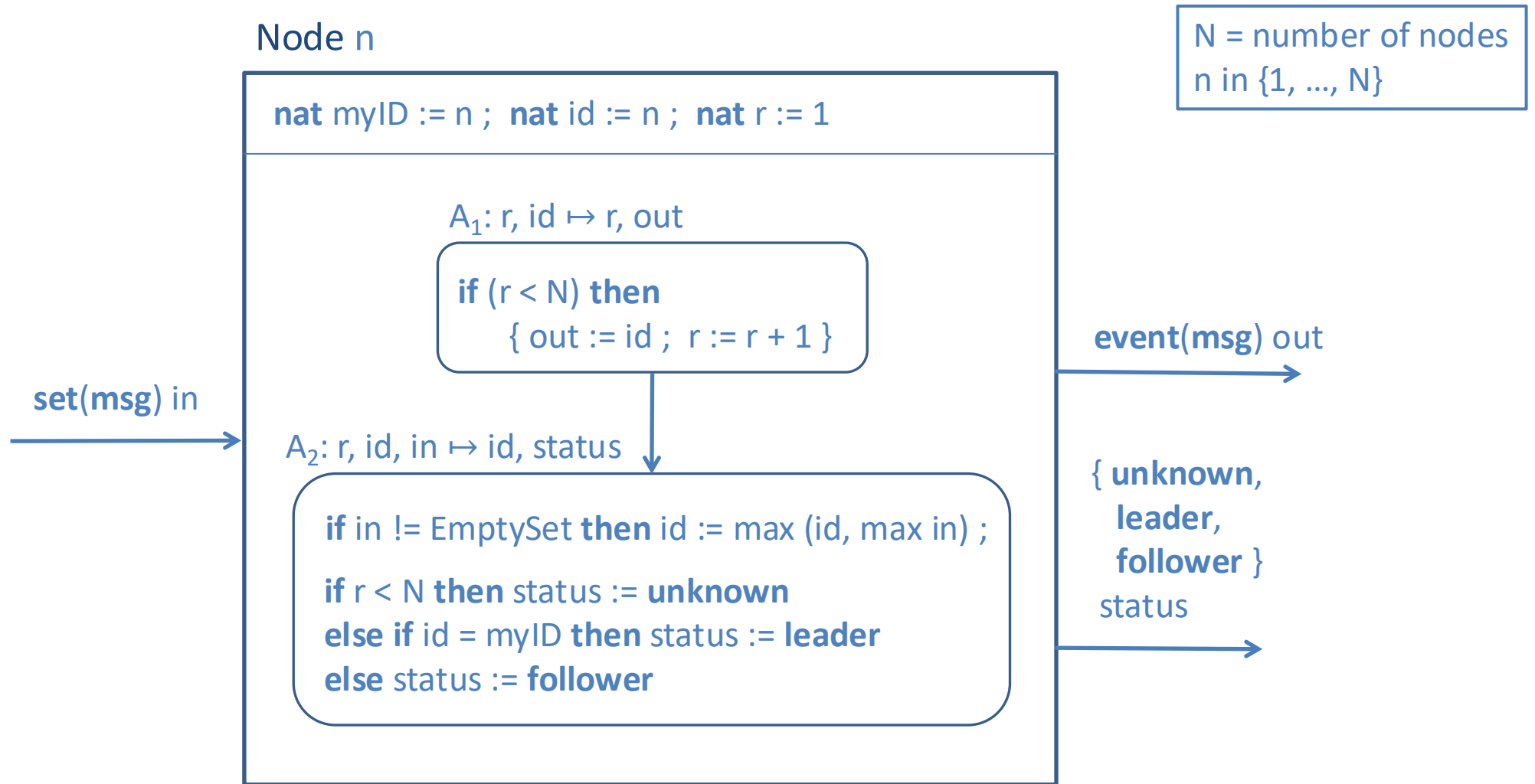


Third round

Leader Election

- When should a node stop and make a decision?
 - When it knows that enough rounds have elapsed for message from every node to reach every other node
- Correctness depends on following assumptions:
 1. Network is *strongly connected*: for all nodes m and n , there is a directed path from m to node n
 2. All nodes know an *upper bound* N on total number of nodes
- Implementation of decision rule:
 - Maintain a state variable r to count rounds, initially 1
 - In each round, r is incremented
 - When $r = N$, decide
- What should the decision be?

Node Component for Leader Election



Leader Election

General questions:

- Does a node really have to wait for N rounds?
- If a node receives a value higher than its own identifier, can it stop participating (i.e. transmit no more messages)?
- Does a node have to transmit in each round?
When can it choose to skip a round without affecting correctness?

Credits

Notes based on Chapter 2 of

[Principles of Cyber-Physical Systems](#)

by Rajeev Alur

MIT Press, 2015